LAKEHOUSE ASSESSMENT

Enterprise Lakehouse Advisory Services

Three services · Health Check, Modernization Assessment, Technology Deep Dive

1 **Why it matters**
What an agent or a report can answer depends on the lakehouse underneath it. p. 2

2 **Three services**
Health Check, Modernization Assessment and Technology Deep Dive — what each covers, how long it takes. p. 3 – 6

3 **How it is run**
One method, four kinds of evidence, phases and team by service. p. 7

4 **What comes out and next steps**
Deliverables by service, the follow-on work, what is needed from your side. Appendix from p. 10. p. 8 – 9

Where It Sits — One Question, Four Layers

KOMTAS

An agent's answer is built from the layers below it. These services assess the bottom one: the lakehouse.

01 · WHY IT MATTERS

"How many active customers churned last week, by region?"

"1,284 customers; most in the Aegean region."
returned with its definition, source table and freshness

THESE SERVICES

Action 360
Intelligent Work Experience

Where the question is asked and the answer lands: Teams, Slack, portal, API.

Agent 360
Governed Digital Workforce

Turns the question into a query and acts on the result.

Context 360
Enterprise Context

Says what "active customer" and "churn" mean: glossary, ontology, semantic layer.

Data / Analytic 360
Trusted Data Foundation

Holds the data: lakehouse tables, pipelines, catalogue, lineage and access rules.

The answer travels back up — it is only as good as the bottom layer

WHERE A WEAK LAKEHOUSE SHOWS UP IN THE ANSWER

Millions of small files, no compaction

Queries slow down; the answer comes late or times out.

Storage & data architecture

Late events were never reprocessed

The answer is incomplete and does not say so.

Data integration & processing

Tables are catalogued but have no owner

Nobody can confirm which table is the right source.

Governance & security

Clusters never scale down

The answer is right, but each one costs too much.

Cost visibility

Three Services — Which One Fits

KOMTAS

Enterprise Lakehouse Advisory Services. Each can be taken on its own; none is a mandatory step before another.

02 · THREE SERVICES

p. 4

A Health Check

Assess current state · technology-agnostic

Focus	High-level view of the current architecture
Areas	6 areas, technology-agnostic
Result	Red / amber / green per area
Access	None — guided
Evidence	Statement · Document · Observation · Measurement
Weeks	S 1 · **M 1.5** · L 2

p. 5

B Modernization Assessment

Define future state · current state included

Focus	Current state, target architecture and roadmap
Areas	8 areas, current and future
Result	Current and target level on a 5-level ladder
Access	Guided or read-only
Evidence	Statement · Document · Observation · Measurement
Weeks	S 3 · **M 3.5** · L 4

p. 6

C Technology Deep Dive

Optional specialist assessment · one technology

Focus	One technology, one technical problem
Areas	Up to 7 areas, chosen together
Result	Full measurement on the platform's own numbers
Access	Read-only required
Evidence	Statement · Document · Observation · Measurement
Weeks	S 2 · **M 3** · L 4

WHICH ONE FITS · THE QUESTION ON THE TABLE DECIDES

- Is an independent view of the main risks needed, quickly and before a decision? → A
- Is a target architecture and a sequenced modernisation roadmap needed? → B
- Is there one technology with one hard problem — performance, cost or configuration? → C

A · Enterprise Lakehouse Health Check

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Assess current state · technology-agnostic.

02 · THREE SERVICES · A

A

Health Check

High-level view of the current architecture

WEEKS S · M · L	KOMTAŞ EFFORT person-days	ACCESS
1 · 1.5 · 2	6 · 9 · 12	None — guided

WHAT IS ASSESSED · 6 AREAS

1. **Architecture & Technology Landscape**
   Components, layers, storage and table format choices, environments
2. **Data Integration & Processing**
   Batch and streaming pipelines, ingestion patterns, failure handling
3. **Governance & Security**
   Catalogue, ownership, lineage, quality rules, access control
4. **Performance & Scalability**
   Query behaviour, concurrency, growth headroom
5. **Platform Operations**
   Orchestration, monitoring, incident handling and recovery
6. **Cost Visibility**
   Where the spend goes and who it can be attributed to

HOW IT IS ASSESSED

Red / amber / green per area

Red | Amber | Green

From documents, interviews and walkthroughs. No score is produced.

WHAT YOU GET

- Red / amber / green view of six areas
- Findings and key risks, each with its evidence
- Improvement recommendations, prioritised
- Advice on whether B or C is worth taking

Not a product audit. The aim is to understand the architecture and its main risks without going into product internals.

B · Enterprise Lakehouse Modernization Assessment

Define future state · current state included.

02 · THREE SERVICES · B

B

Modernization Assessment

Current state, target architecture and roadmap

WEEKS S · M · L	KOMTAŞ EFFORT person-days	ACCESS
3 · 3.5 · 4	22 · 28 · 34	Guided or read-only

WHAT IS ASSESSED · 8 AREAS

1. **Current & Future-State Architecture**
   What to keep, what to redesign; the target lakehouse architecture
2. **Data Integration Modernization**
   Batch and streaming, CDC, pipeline patterns and late data
3. **Query & Analytics Enablement**
   Engines, serving to BI, concurrency and freshness commitments
4. **ML & Feature Engineering Enablement**
   Data supply and feature production for ML — not MLOps
5. **Governance & Security Modernization**
   Catalogue, lineage, RBAC, data quality rules
6. **Platform Operations & Observability**
   Monitoring, alerting, orchestration, recovery
7. **Platform Scalability & Cost**
   Growth, compute separation, cost attribution
8. **Technology Evolution**
   Open formats, engine independence, components to adopt or retire

HOW IT IS ASSESSED

Current and target level per area



Five-level ladder, placed on evidence. No weighted score; the gap per area drives the roadmap.

WHAT YOU GET

- Current state analysis on eight areas
- Target lakehouse architecture: keep, redesign, change or add serving
- Prioritised modernisation needs
- Implementation roadmap with sequence and effort

ML is covered as data supply and feature engineering. Model development and MLOps are outside the scope.

C · Technology-Specific Deep Dive

Optional specialist assessment · one technology.

02 · THREE SERVICES · C

C

Technology Deep Dive

One technology, one technical problem

WEEKS by areas 1–2 · 3–4 · 5–7	KOMTAŞ EFFORT person-days	ACCESS
2 · 3 · 4	12 · 18 · 26	Read-only required

WHAT IS ASSESSED · 7 AREAS · MENU, CHOSEN IN P0

1. **Storage & Data Architecture**
   File layout, partitioning, compaction, snapshots and retention
2. **Query & Processing Performance**
   Query profiles, p95, skew, caching, job run times
3. **Data Integration & Pipeline Engineering**
   Job design, idempotency, late data, reprocessing
4. **Data Governance & Security Configuration**
   Policies, RBAC / ABAC, encryption, audit trail
5. **Platform Operations & Observability**
   Metrics, alerts, runbooks, failure response
6. **Infrastructure & Resource Optimization**
   Cluster sizing, autoscaling, idle compute, storage tiers
7. **Platform Configuration & Reliability**
   Configuration review, HA / DR, upgrade and patch path

Areas outside Komtaş delivery experience are run with a named specialist, or left out of scope.

HOW IT IS ASSESSED

Full measurement, selected areas

The platform's own numbers over an agreed window: logs, metadata, metrics

Each finding is tied to a number. Nothing is estimated where it can be measured.

WHAT YOU GET

- Measurements on the selected areas
- Technical findings, each tied to a number
- Configuration and tuning recommendations
- Effort estimate for the fixes

Areas are chosen in P0. Where Komtaş has no delivery experience on the technology, a named specialist joins and is shown in the report.

How It Is Run — One Method, Three Depths

KOMTAS

The same evidence types and phases are used in every service. The service sets how deep the evidence goes and how long each phase takes.

03 · HOW IT IS RUN

FOUR KINDS OF EVIDENCE

Measurement — B · C

A number from the platform over a stated window

Document — A · B · C

A configuration or document that exists on its own

Observation — A · B · C

Seen working, or failing, in a walkthrough

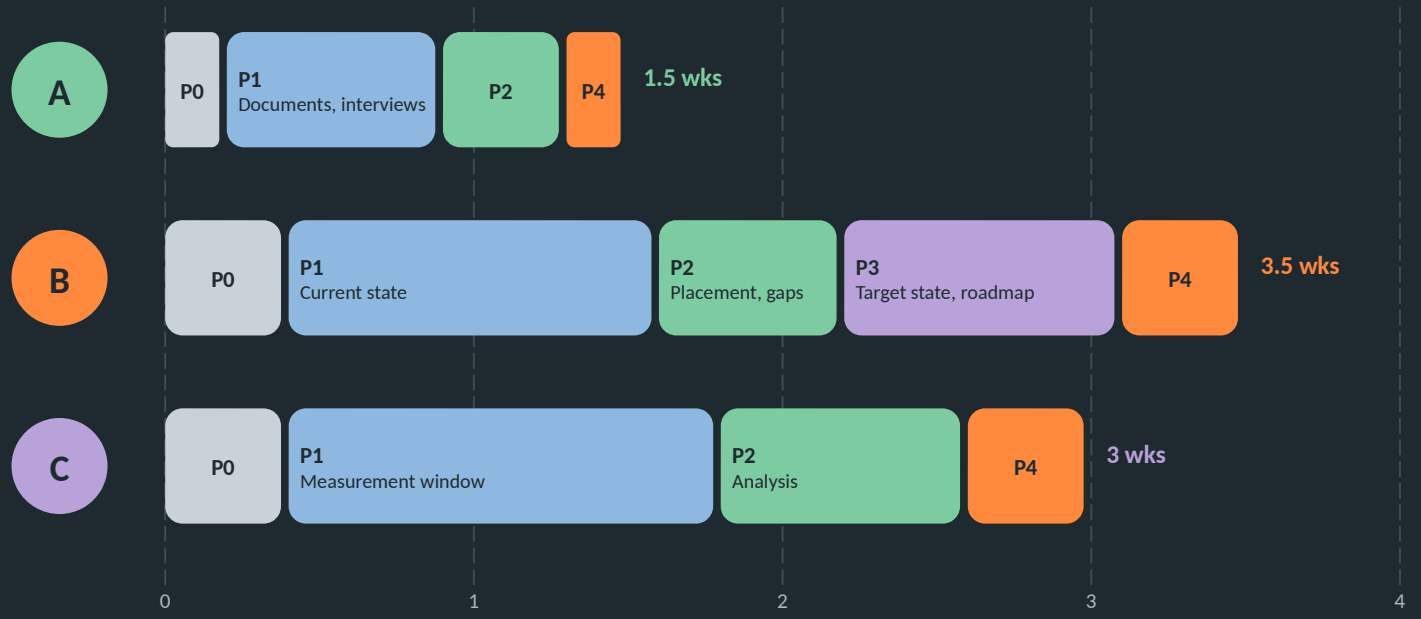
Statement — A · B · C

What a person says — recorded as a claim

A level is placed only where a document, an observation or a measurement supports it. Statements alone are reported as claims.

Expertise is stated, not assumed. Deep technical work is committed only where Komtaş has delivery experience on the technology, or with a named specialist.

PHASES BY SERVICE · M BAND, IN WEEKS



P0 scope · P1 evidence · P2 placement or analysis · P3 target state and roadmap (B only) · P4 read-out

WHO DOES THE WORK

- A — Assessment lead · data architect
- B — Lead · architect · data engineer · governance and BI / ML on call
- C — Platform specialist for the technology · lead for scope and report

What Comes Out — and Where It Leads

KOMTAS

What each service hands over, and the follow-on work an assessment usually opens.

04 · WHAT COMES OUT

WHAT IS HANDED OVER

	A	B	C
Findings and key risks, each with its evidence	✓	✓	✓
Red / amber / green per area	✓	–	–
Current and target level per area	–	✓	–
Target lakehouse architecture — keep, redesign, change, add serving	–	✓	–
Prioritised modernisation needs and roadmap	–	✓	–
Measurements and tuning recommendations	–	–	✓
Dated baseline for the next assessment	✓	✓	✓

WHERE IT USUALLY LEADS

- **Architecture improvement** — fixes inside the current platform
- **Modernization** — Data Modernization, proposed separately
- **Data integration** — pipelines, CDC, streaming
- **Technology investment** — components to add, adopt or retire
- **Managed Services** — running and monitoring the platform

Each service ends with a dated baseline. The same areas and definitions are used again 6–12 months later, so improvement is shown, not claimed. A Health Check can be followed by B or C; evidence already collected is reused and not billed twice.

What We Need From You — and the Next Steps

KOMTAS

Three roles on your side, two constraints that set the date, and four steps from today to the start of the work.

04 · NEXT STEPS

WHO WE NEED ON YOUR SIDE

Sponsor

start and read-out

Sets the decision the work supports and the service to take. Clears approvals.

Platform owner

throughout · start condition

Knows the environment, explains how it runs, checks findings before they are issued.

Data engineering team

evidence phase

Explains jobs and pipelines and joins the walkthroughs.

Security and infrastructure open the approvals in B and C; consumer representatives join the interviews.

TWO THINGS THAT SET THE DATE

Access

A needs none. B runs guided or with read-only access, agreed at the start. C needs read-only access to the technology.

Each source is a separate approval

Log and metadata retention

How many days of query, job and snapshot history the platform keeps — matters for B and C.

Under 30 days adds an observation window

Neither can be shortened by adding people. Both are asked in the short form.

NEXT STEPS

1. **Service walkthrough** — today
   you are here
2. **Service choice and short form** — about one hour
3. **Proposal** — service, band, schedule
4. **Kick-off** — scope, access, definitions

Start where the question is. A Health Check in one to two weeks, a Modernization Assessment in three to four, or a Deep Dive on the one technology that needs it.

APPENDIX

Method in detail — for reference and questions

A

Scope and context

Where the services sit · What is and is not assessed · Seven recurring lakehouse patterns

p. 11 – 13

B

Method

Maturity ladder · Evidence model · Roles · Duration and effort

p. 14 – 17

C

Decisions and limits

Target state decision (Modernization) · Limits of the assessment

p. 18 – 19

Where This Assessment Sits — AI Value Proposition

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Four layers. Each layer consumes the one below. This work measures the bottom layer — it does not build it.

Action 360

Intelligent Work Experience

Teams | Slack | WhatsApp | Web / Portal | Claude Desktop | API, Headless

Agent 360

Governed Digital Workforce

INDUSTRY: Manufacturing | Retail | Banking | Energy | Aviation | Other

FUNCTION: Customer Service | Sales | Finance | Procurement | Operations | Other

Context 360

Enterprise Context

Semantic Layer | Ontology | Knowledge Graph | Master Data | Corporate Memory

Data / Analytic 360

Trusted Data Foundation

THIS WORK · MEASUREMENT AND ROADMAP

Lakehouse | Datawarehouse | Integration | Data Virtualization | Governance / MDM | Access Control

The assessment measures this layer; it does not build it. Build and migration fall under Data Modernization and are proposed separately, after the assessment.

RELATION TO ADJACENT LAYERS

Relation to the DWH

If both exist, the one carrying most analytical load is assessed; the other only at its interface.

Relation to Context 360

Semantic layer and ontology feed from this layer. If the lakehouse is weak, the layer above inherits the errors.

Relation to Agent 360

An agent is only as reliable as the freshness and accuracy of the data it works on.

What It Assesses, What It Does Not

KOMTAS

The subject is the lakehouse itself; adjacent layers are covered only through their interfaces

ASSESSED

- Object storage layout, file size and partition design
- Table format usage — transaction, snapshot, retention discipline
- Ingestion and streaming pipelines, late-arriving data behaviour
- Processing layer: re-runnability, dependencies, idempotency
- Catalogue, ownership, lineage and data quality rules
- Query engine behaviour, concurrency and cost
- Access control, encryption and audit trail
- Lakehouse side of ML feeding and the BI interface

NOT ASSESSED

- Source systems' own architecture and data quality
- The data warehouse's own capability — a separate assessment
- BI tools' report inventory, adoption and licensing
- Application layer, APIs and operational systems
- Master data management and golden record design
- Agent and model development processes
- Cloud contracts, network and hardware
- Build, migration and implementation of improvements

LAYER BOUNDARY IS KEPT

Adjacent layers are assessed only through their interfaces with the lakehouse; they are not measured on their own.
Without this boundary the assessment spreads across the whole data architecture, the timeline triples and no dimension is examined in enough depth.

MEASURES, DOES NOT BUILD

This work produces findings and a roadmap; it does not implement improvements. Implementation is a separate engagement, proposed separately.
Without this separation the assessment becomes the report of a party implementing its own recommendations and loses its independence.

NOT PRODUCT CHOICE

The assessment is vendor-independent. The output is not “which product to buy”, but how much of the existing investment can be kept, what needs to be redesigned, and which evidence that decision rests on.

Seven Recurring Patterns in Lakehouse Implementations

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Each is measurable; the assessment is not done to spot them but to show them with evidence

Lake plus always-on cluster

Storage and compute are said to be separated, but the cluster never shuts down. The lakehouse cost promise is lost here.

Measured: cluster idle ratio, autoscaling event count, cost per query

Small file explosion

Millions of small files pile up; every query is spent reading metadata. The most common and easiest-to-measure degradation.

Measured: average file size, files per partition, compaction frequency

Over-partitioning

Every column gets a partition; data per partition drops to a few megabytes and the planner slows down.

Measured: partitions per table, average partition size, partitions scanned per query

Table format in place, but bypassed

Iceberg or Delta is installed, but some pipelines write parquet directly. The transaction guarantee is silently lost.

Measured: share of tables under table format, orphan file ratio, writes outside the catalogue

Full catalogue, no owner

Tables are catalogued but have no defined owner. No one dares to delete them, and storage keeps growing.

Measured: share of tables with an owner, last access date, share of unused tables

Open format, closed engine

The format is open, but proprietary extensions mean only one engine can read the tables. The portability promise does not hold.

Measured: whether a second engine can read the tables, use of proprietary features

Layer names, no rules

Bronze, silver, gold naming exists, but there is no written rule defining which data sits in which layer.

Measured: tables holding the same data in more than one layer, existence of a layer promotion rule

ALL SEVEN LOOK TECHNICAL; FIVE ARE GOVERNANCE ISSUES

Small files, over-partitioning and a bypassed table format are not design errors but the result of missing written rules and ownership.
The roadmap therefore covers not only technical fixes but also which rule is to be owned by whom.

Five-Level Maturity Ladder

KOMTAS

Each dimension is placed on this ladder. A level is not claimed — it is shown with evidence; without evidence, one level lower is recorded.

LEVEL 1
Ad hoc

No rules; results depend on people

- No written standard; practices vary by team
- Problems are solved by improvisation
- No measurement; status unknown

Exit: practice is written down

LEVEL 2
Repeatable

Work is done the same way, but not written

- Experienced staff follow the same method
- A newcomer can't repeat the result
- Measurement, if any, is irregular

Exit: rule is documented and owned

LEVEL 3
Defined

Rule is written and applied

- The standard is documented and known
- Deviations are noticed
- Regular measurement, no thresholds

Exit: thresholds set, deviation measured

LEVEL 4
Managed

Measured, thresholds set, deviation visible

- Thresholds defined and monitored
- Deviation detected automatically
- Root cause analysis is done

Exit: correction is automated

LEVEL 5
Optimised

Deviation auto-corrected; continuous improvement

- Correction is largely automated
- Thresholds are updated based on data
- Improvement shown by measurement

TARGET LEVEL IS THE ORG'S DECISION

Moving every dimension to level five is not the goal. The target level is set per dimension by the organisation; the assessment measures the gap between current and target. A level the organisation cannot show a need for is a cost, even if reached.

NO LEVEL SKIPPING

Level four cannot be reached without level three: setting a threshold first requires a written rule.
The roadmap is therefore sequential; a plan that skips two levels at once is not feasible.

Evidence Model

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Each finding is tied to an evidence type, shown in the report. If the evidence type is hidden, the report is unreadable.

MEASUREMENT
Measurement

A number the platform produces

· Query logs, job history, metadata statistics, billing data

Highest confidence · not open to dispute

ARTEFACT
Document

An existing document or configuration

· Table definition, policy file, architecture document, access configuration

High confidence · currency is verified

OBSERVATION
Observation

A flow traced end to end

· Tracing a load, following a report back to source, a recovery drill

Medium confidence · one-off observation

STATEMENT
Statement

What people report

· Interviews, surveys, team statements

Low confidence · cannot confirm a level alone

THREE-STATE EVIDENCE SCALE

Evidence provided
Level shown by measurement or document

Partial evidence
Evidence exists, but its scope is narrow or its currency unclear

No evidence
Could not be measured due to access, retention period or missing records

STATEMENT ALONE DOES NOT CONFIRM A LEVEL

Stakeholder statements are a valuable input; however, a dimension resting only on statements is flagged as such in the report.
Otherwise a measured dimension and a narrated one are read with the same confidence, and the report cannot be used for decisions.

DIMENSION WITHOUT EVIDENCE NOT HIDDEN

A dimension that cannot be measured due to short log retention is not guessed as “good” or “bad”; it is recorded as not measurable.
This finding is often an improvement item in itself: what cannot be measured cannot be managed.

Competencies and Roles

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Which role is active in which phase, including expected customer participation

ROLE	DUTIES	PARTY	ACTIVE PHASES
Assessment Lead	Applying the method, scope, decision gates, report	Komtaş	P0 – P4
Data Architect	Architecture and design dimensions, target state decision	Komtaş	P0, P2 – P4
Lakehouse / Platform Engineer	Storage, table format, partition, engine and performance measurements	Komtaş	P1 – P3
Data Engineer	Review of ingestion, processing and data quality pipelines	Komtaş	P1 – P2
Governance Consultant	Catalogue, lineage, ownership and metadata dimensions	Komtaş	P1 – P3
Security and Privacy Specialist	Access control, encryption, audit trail	Komtaş	P1 – P2
FinOps / Cost Analyst	Compute and storage cost, attributability	Komtaş	P1 – P2
BI / Analytics Consultant	Serving interface and consumption-side dimensions	Komtaş	P1 – P2
Project sponsor	Decision-making, prioritisation, raising access approvals	Customer	P0, phase transitions
Platform owner	Environment information, configuration access, validation of findings	Customer	P0 – P4
Data Engineering Team	Job and pipeline information, end-to-end tracing support	Customer	P1
Security & Infrastructure	Access approvals, policy and configuration information	Customer	P0 – P1
Consumer Representatives	Expectations, freshness needs and issues experienced	Customer	P1

NO ASSESSMENT WITHOUT A PLATFORM OWNER

If no one with access to configuration and environment information is assigned, P1 extends and most evidence falls back to statements.
Assigning this role is a precondition to start.

Duration and Effort by Service

KOMTAŞ

Indicative band values. The band is set from the short form; dates are fixed at kick-off, once access lead times are known.

SERVICE	SMALL	MEDIUM	LARGE
A · Health Check 6 areas · RAG	1 week · 6 pd · 2 days	1.5 weeks · 9 pd · 3 days	2 weeks · 12 pd · 4 days
B · Modernization Assessment 8 areas · ladder	3 weeks · 22 pd · 5 days	3.5 weeks · 28 pd · 6 days	4 weeks · 34 pd · 8 days
C · Technology Deep Dive by number of areas	1–2 areas · 2 weeks · 12 pd	3–4 areas · 3 weeks · 18 pd	5–7 areas · 4 weeks · 26 pd

Cell format: duration · Komtaş person-days · customer person-days (C: areas · duration · Komtaş person-days). Specialist effort in C is quoted separately.

SIZING DRIVERS

Number of tables	≤ 300	300 – 1,500	> 1,500
Source systems	≤ 5	5 – 15	> 15
Streaming pipelines	0 – 2	3 – 10	> 10
Query engines / table formats	1	2	> 2
Environments	≤ 2	3	> 3
Access approval time (B, C)	≤ 2 weeks	2 – 6 weeks	> 6 weeks

BAND RULE · the highest band that at least three drivers fall into. One large driver alone does not set the band.

Target State Decision

KOMTAS

Modernization (B) only. Four options. The decision is made on decision criteria and the findings behind them, not on a score.

OPTION A Keep and optimise

Current structure is sound; issues are operational

- Table format is used with discipline
- Partition and file layout reasonable
- Issues are at compaction, statistics and monitoring level

OPTION B Redesign in place

Structure right, design decisions wrong

- Table format exists but is bypassed
- Partition design or file layout is broken
- Catalogue and ownership missing, but no platform change needed

OPTION C Change engine or platform

Current setup cannot carry the target

- Engine does not meet concurrency or scale needs
- Proprietary dependency blocks portability
- Cost model does not fit the workload

OPTION D Add a serving layer

Lakehouse is right; consumption side is strained

- Analytical load fits; BI concurrency is insufficient
- Freshness commitment cannot be met by the query engine
- Existing investment can be kept

CRITERION	A KEEP	B REDESIGN	C CHANGE	D SERVING
Is the table format used with discipline	STRONG	WEAK	FITS	STRONG
Is file and partition layout reasonable	STRONG	WEAK	FITS	STRONG
Does the engine meet scale and concurrency	STRONG	FITS	WEAK	CONDITIONAL
Does proprietary dependency block portability	FITS	FITS	STRONG	FITS
Are BI freshness and concurrency sufficient	STRONG	FITS	FITS	WEAK
Does the cost model fit the workload	STRONG	FITS	WEAK	CONDITIONAL

READING CRITERIA

"WEAK" in a column means the option loses validity when the answer to that criterion is negative.
Rejected options are covered in the report in the same detail as the recommended one, with the conditions for reconsidering them.

Limits of the Assessment

KOMTAS

What this work does not show is stated explicitly within the report

No load testing or capacity planning

The performance dimension is based on measuring the current workload. Whether a future load can be carried is a separate study.

No code-level security audit

Access policies and configuration are reviewed; vulnerabilities in processing code are not in the scope of this assessment.

Cost figures are not negotiated

The cost dimension measures current consumption and attributability. Cloud contracts and discount structures are out of scope.

Behaviour outside the window is not covered

Measurements cover the agreed period. Unusual behaviour outside that period is not seen, and this is stated in the report.

No “pass” result is produced

The assessment is not a certification. It shows where the platform stands, on what evidence, and what to do next.

No improvement without implementation

The report produces a roadmap, not its implementation. Implementation is separate work, proposed separately.

LIMITS ARE STATED IN THE REPORT

Areas left out of scope are collected in one section of the report. “No finding in this dimension” and “this dimension was not reviewed” are not the same; a report without this distinction creates false confidence in areas not reviewed.

A DIMENSION WITHOUT EVIDENCE IS ALSO A FINDING

A dimension that cannot be measured due to log retention, access restrictions or missing records is often an improvement item in itself. The report states that what cannot be measured cannot be managed.

OBJECTIONS ARE RECORDED, NOT RESOLVED

If the platform team disagrees with a finding, the objection stays in the report next to the finding with its rationale; the discussion is not removed.